

ECEN-662: Homework 7

Due on April 12, 2026 at 11:59 PM

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Problem 1

Let $X = (X_1, \dots, X_n)^t$ where

$$X_i \stackrel{iid}{\sim} \text{Bernoulli}(\theta),$$

where θ is the unknown parameter. Assume that the prior distribution for Θ is $\mathcal{U}((0, 1))$.

- (i) Find the Bayes estimator of θ under the squared-error loss.
- (ii) Now let X_{n+1} be a new observation such that $X_{n+1} \sim \text{Bernoulli}(\theta)$ and is independent of X given θ . Find $\mathbb{P}(X_{n+1} = 1|X)$.

Solution

Problem 2

Suppose that X has a density of exponential form

$$f_{\theta}(x) \propto_x h(x) \exp \left(\sum_{i=1}^d \theta_i T_i(x) - A(\theta) \right)$$

where each $T_i(x)$ is differentiable.

- (i) Show that if X has support in real line and g is any differentiable function such that $\mathbb{E}_{\theta}[g'(X)] < \infty$, then

$$\mathbb{E}_{\theta} \left[\left(\frac{h'(X)}{h(X)} + \sum_{i=1}^d \theta_i T'_i(X) \right) g(X) \right] = -\mathbb{E}_{\theta}[g'(X)].$$

- (ii) Using part a), show that if $X \sim \mathcal{N}(\mu, \sigma^2)$, then

$$\text{Cov}_{\theta}[g(X), X] = \sigma^2 \mathbb{E}_{\theta}[g'(X)]$$

where $\theta = [\mu, \sigma^2]^t$.

- (iii) Use part ii) to compute the third and fourth moments of $\mathcal{N}(\mu, \sigma^2)$.

Solution

Problem 3

Let

$$f_{\theta}(x) = h(x) \exp(\theta x - A(\theta))$$

be a one-parameter, canonical exponential family.

(i) Show that for any prior $\pi(\theta)$,

$$\mathbb{E}_{\pi}[\Theta \mid X = x] = \frac{d}{dx} \log \left(\frac{m_{\pi}(x)}{h(x)} \right),$$

where $m_{\pi}(x)$ is the marginal distribution of X .

(ii) Use part i) to prove Tweedie's formula:

$$\mathbb{E}_{\pi}[\Theta \mid X = x] = x + \frac{d}{dx} \log m_{\pi}(x),$$

when f_{θ} is $\mathcal{N}(\theta, 1)$.

Solution

Problem 4

Assume that

$$\Theta_i \stackrel{iid}{\sim} \pi(\theta_i) \quad \text{and} \quad X_i \mid \Theta_i = \theta_i \stackrel{ind}{\sim} \text{Poisson}(\theta_i),$$

Show that for any prior π and any $i \in [n]$, we have

$$\mathbb{E}_\pi [\Theta_i \mid X_i = x_i] = \frac{(x_i + 1) m_\pi(x_i + 1)}{m_\pi(x_i)} = \frac{(x_i + 1) \mathbb{E}_\pi [Y_{x_i+1}]}{\mathbb{E}_\pi [Y_{x_i}]},$$

where $Y_x = \sum_{i=1}^n 1_{\{X_i=x\}}$.

Solution